Let V be a fin. dim vector space, and $T: V \to V$ be a linear map.
Define $\chi_T(t) = \det(tI - [T]_\beta)$ be the characteristic polynomial
$m_T(t)$ be the minimal polynomial.

Lemma. For $c \in F$,

$$\chi_T(c) = 0 \text{ if and only if } \ker(T - cI) \neq \{0\}.$$

Proof. $\chi_T(c) = 0 \Leftrightarrow \det(cI - [T]_\beta) = 0 \Leftrightarrow cI - [T]_\beta$ is non-singular.

Lemma. For $c \in F$.

$$\chi_T(c) = 0 \text{ if and only if } m_T(c) = 0.$$

Proof. Suppose that $\chi_T(c) = 0$. Then, $T(\alpha) = c \cdot \alpha$ for some non-zero $\alpha \in V$.
Therefore, $0 = m_T(T)(\alpha) = m_T(c) \cdot \alpha$, but since $\alpha \neq 0$, $m_T(c) = 0$.
Conversely, suppose that $m_T(c) = 0$. Then,

$$m_T(t) = (t - c) \cdot g(t) \text{ for some } g(t) \in F[t].$$

By the minimality of m_T, $g(T) \neq 0$, so that there exists $\alpha \in V$
such that $g(T)(\alpha) \neq 0$, thus

$$(T - cI)(g(T)(\alpha)) = m_T(T)(\alpha) = 0$$

So proved. ☑

「The Primary decomposition theorem.
Let T be a linear operator on the finite-dimensional vector space V over a field F.
Let P be the minimal polynomial for T with the irreducible decomposition

$$P = p_1^{r_1} \cdot p_2^{r_2} \cdots p_k^{r_k} = \prod_{i=1}^{k} p_i(x)^{r_i}.$$

Then,

$$V = \bigoplus_{i=1}^{k} \ker p_i^{r_i}(T) \quad \text{and each } \ker p_i^{r_i}(T) \text{ is invariant } T.$$

Moreover, the restriction of T by $\ker p_i^{r_i}(T)$ has the minimal polynomial as $p_i^{r_i}$.

「Thm. Let α be any non-zero vector in V and let p_α be the T-annihilator of α.
1) $\deg p_\alpha = \dim Z(\alpha, T)$
2) $Z(\alpha, T) = \text{span}\{T^k\alpha \mid 0 \le k \le \deg p_\alpha\}$
3) The minimal polynomial of $T|_{Z(\alpha,T)}$ is p_α.

「Cyclic decomposition Theorem.
「Thm. Let $T: V \to V$ be a linear operator on finite-dimensional space V.
Suppose that the minimal polynomial of T has a irreducible decomposition

$$m_T(x) = p_1(x)^{r_1} \cdots p_k^{r_k}$$

Then, by the primary decomposition, V is decomposed by T-invariant spaces

$$V = \ker p_1^{r_1} \oplus \cdots \oplus \ker p_k^{r_k}$$

With $T|_{\ker p_i^{r_i}}$ has minimal polynomial as $p_i^{r_i}$ 이것까지가 지난 복습.

Now, for fixed $1 \le s \le k$, denote $T_s = T|_{\ker p_s^{r_s}}$ and $W_s = \ker p_s^{r_s}$.
Then, there are non-zero vectors $\alpha_1, \cdots, \alpha_n$ such that

$$W_s = Z(\alpha_1, T_s) \oplus \cdots \oplus Z(\alpha_n, T_s)$$

satisfying for some unique integers $1 \le r_n \le \cdots \le r_2 \le r_1 = r_s$,

$$m_{T_s|Z(\alpha_s, T_s)} = p_s^{r_s}.$$

Primary decomposition: splits case.

Suppose that the minimal polynomial of T splits as

$$m_T(t) = (t-\lambda_1)^{r_1} \cdots (t-\lambda_k)^{r_k}.$$

Then, by the Primary decomposition,

$$V = \ker(T-\lambda_1)^{r_1} \oplus \cdots \oplus \ker(T-\lambda_k)^{r_k}.$$

Now, we get the resolution of identity

$T = T\circ E_1 + \cdots + T\circ E_k$ where E_i is projection onto $\ker(T-\lambda_i)^{r_i}$ along $\bigoplus_{j\neq i} \ker(T-\lambda_j)^{r_j}$.

and let

$$D = \lambda_1 E_1 + \cdots + \lambda_k E_k.$$

Then,

$$N = T - D = (T-\lambda_1 I)\circ E_1 + \cdots + (T-\lambda_k I)\circ E_k.$$

is nilpotent, with the order is at most $\max\{r_1, \cdots, r_k\}$.

Moreover, for $i \neq j$,

$$(T-\lambda_i I)\big|_{\ker(T-\lambda_j)^{r_j}} = \underbrace{(\lambda_j - \lambda_i)\cdot I}_{\text{invertible}} + \underbrace{(T-\lambda_j I)\big|_{\ker(T-\lambda_j)^{r_j}}}_{\text{nilpotent}}$$

So it is invertible.

Jordan form.

Let $T: V \to V$ be a linear operator on n-dimensional vector space over F, with $\chi_T(t) = (t-\lambda_1)^{r_1} \cdots (t-\lambda_k)^{r_k}$

and $m_T(t) = (t-\lambda_1)^{s_1} \cdots (t-\lambda_k)^{s_k}$ $(1 \le s_i \le r_i, \forall i.)$

1). $\bigcup_{N=1}^{\infty} \ker(T-\lambda_i)^N = \ker(T-\lambda_i)^{s_i}$.

Proof. The $\supseteq$ inclusion is clear. And, by the primary decomposition, given $x \in \bigcup_{N=1}^{\infty} \ker(T-\lambda_i)^N$, there is unique representation

$$x = x_1 + \cdots + x_k \quad \text{where } x_j \in \ker(T-\lambda_j)^{s_j}, \ \forall j.$$

Fix $m \in \mathbb{N}$ such that $x \in \ker(T-\lambda_i)^m$. That is,

$$(T-\lambda_i I)^m(x) = \sum_{j=1}^{k} (T-\lambda_i I)^m(x_j) = 0.$$

Meanwhile, each W_j is $T-\lambda_i I$ invariant and $(T-\lambda_i I)|_{\ker(T-\lambda_j I)^{s_j}}$ is invertible, therefore $x_j = 0$ for all $j \neq i$. Finally, $x = x_i \in \ker(T-\lambda_i)^{s_i}$, So Proved. □

(구해지는 것은, 각각이 independant.)

2). $\dim \ker(T-\lambda_i)^{s_i} = \dim \bigcup_{N=1}^{\infty} \ker(T-\lambda_i)^N = r_i$.

Proof. By the primary decomposition, $V = \oplus \ker(T-\lambda_i)^{s_i}$, So that

$$\chi_T(t) = \prod_{i=1}^{k} \chi_{T|_{\ker(T-\lambda_i)^{s_i}}}(t) = \prod_{i=1}^{k} (t-\lambda_i)^{\dim \ker(T-\lambda_i)^{s_i}}$$

, So Proved. □

(In $\ker(T-\lambda_i)^{s_i}$, it has only λ_i as eigenvalue.)

3). s_i = largest Jordan block size for λ_i.

dim $\ker(T-\lambda_i I)$ = number of λ_i-blocks.

proof. $J_{n_1}(\lambda) \oplus \cdots \oplus J_{n_g}(\lambda)$, then $m_{T|\lambda}(t) = \mathrm{lcm}((t-\lambda)^{n_1}, \cdots,) = (t-\lambda)^{\max n_j}$.

Summary.

Jordan form.

Let $T: V \to V$ be a linear operator on n-dimensional vector space over F, with $\chi_T(t) = (t-\lambda_1)^{r_1} \cdot \cdots \cdot (t-\lambda_K)^{r_K}$

and $m_T(t) = (t-\lambda_1)^{s_1} \cdot \cdots \cdot (t-\lambda_K)^{s_K}$ $(1 \le s_i \le r_i, \forall i.)$

Then, the Jordan form is: $s_i \ge s_{i2} \ge s_{i3} \ge \cdots$,

$$J = \begin{bmatrix} J_{s_1}(\lambda_1) & & & & \\ & J_{s_{12}}(\lambda_1) & & & \\ & & \ddots & & \\ & & & J_{s_{1n_1}}(\lambda_1) & \\ & & & & J_{s_2}(\lambda_2) \ddots \\ \underbrace{\qquad\qquad}_{r_1} & & & & \end{bmatrix}$$

[Computations.
[HK, 7.3.3,
Let $A \in M_{5\times 5}(\mathbb{C})$ with

$$\chi_A(t)=(t-2)^3\cdot(t+7)^2 \quad \text{and} \quad m_A(t)=(t-2)^2\cdot(t+7).$$

Determine the Jordan form of A.

Proof. Since $(t+7)$ has multiplicity 2 in χ_A and 1 in m_A, the Jordan form of A has $[-7]\oplus[-7]$ as a block corresponding to -7, and $(t-2)$ has multiplicity 2 in m_A, it has a Jordan block

$$J_2(2)=\begin{bmatrix}2&1\\&2\end{bmatrix}$$

and $(t-2)$ has multiplicity 1 in χ_A, the last block is forced $[2]$, Consequently, the Jordan form is

$$J_A=\begin{bmatrix}2&1&&&\\&2&&&\\&&2&&\\&&&-7&\\&&&&-7\end{bmatrix}$$

KAIST. 2021 spring.
Let $A \in M_{8\times 8}(\mathbb{C})$. Suppose $\chi_A(x) = (x+1)^5 \cdot (x-2)^3$

1). Assume
$\dim \ker(A+I) = 2$, $\dim \ker(A+I)^2 = 4$, $\dim \ker(A+I)^3 = 5$
$\dim \ker(A-2I) = 1$, $\dim \ker(A-2I)^2 = 2$, $\dim \ker(A-2I)^3 = 3$.
Determine all possible Jordan canonical form of A.

2). Assume
$$m_A(x) = (x+1)^3 \cdot (x-2)^3.$$
Determine all possible Jordan canonical form of A.

Proof. 1). Since $\dim \ker(A+I)^3 = 5$ equals to the multiplicity of -1 in χ_A,
$\mathbb{R}^8 = \ker(A+I)^3 \oplus \ker(A-2I)^3$. and, moreover,
$\dim \ker(A+I) = 2$ implies A has two Jordan blocks for -1,
$\dim \ker(A-2I) = 1$ implies A has only one Jordan block for 2.
Moreover, $\dim \ker(A+I)^2 - \dim \ker(A+I) = 2$ but
$\dim \ker(A+I)^3 - \dim \ker(A+I)^2 = 1$ therefore it has Jordan blocks of size 2 and 3. Finally, there is only one possible form
$$J_A = J_3(-1) \oplus J_2(-1) \oplus J_3(2).$$

2). Since $(x+1)$ has multiplicity 3 in m_A, there are possible partitions
$5 = 3+2 = 3+1+1$, and since $x-2$ has multiplicity 3 in m_A,
there are two possible forms:
$$J_A = J_3(-1) \oplus J_2(-1) \oplus J_3(2) \quad \text{or}$$
$$= J_3(-1) \oplus [-1] \oplus [-1] \oplus J_3(2)$$

[HK, 7.3.15~16.
Let $N \in M_{n\times n}(\mathbb{C})$ be a nilpotent matrix over complex numbers.
For any non-zero $c \in \mathbb{C}$, $cI+N$ has a square root.
Moreover, deduce that every invertible matrix over $\mathbb{C}$ has a square root.
Proof. Let $A = c\cdot\left(\sum_{k=0}^{n-1} \frac{1}{2^k}\cdot N^k\right)$.